Concentration Units

Mass percent, ppm, mole fraction, molality and molarity

Name _____ Class _____ Date _____

DEFINITIONS AND DATA

c = mol solute / L solution · m = mol solute / kg solvent · mass % = mass solute / mass solution × 100 %

ppm = mass solute / mass solution × 10^6 · x(A) = mol A / total mol

Molar masses (g/mol): NaCl 58.44 · H_2O 18.02 · $C_6H_{12}O_6$ 180.16 · H_2SO_4 98.08

Check which amount goes in the denominator: solution or solvent, volume or mass. Show units in every step.

A. Ideas

1. (a) Explain the difference between molarity and molality. (b) Which concentration units change with temperature, and why?

B. Mass-based units

2. 12.0 g of sugar is dissolved in 188 g of water. Calculate the mass percent of sugar.

3. A 500. g sample of tap water contains 0.45 mg of fluoride ions. Express this in ppm.

4. Saline for a drip is 0.90 % NaCl by mass. What mass of NaCl is in 500. g of saline?

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C. Moles-based units

5. 18.0 g of glucose is dissolved in 90.0 g of water. Calculate (a) the mole fraction of glucose (b) the molality of the solution.

D. Converting between units

6. Concentrated sulfuric acid is 98.0 % H_2SO_4 by mass, with a density of 1.84 g/mL. Calculate its molarity.

7. A NaCl solution has a molality of 1.50 mol/kg. Calculate its mass percent of NaCl.

E. Challenge

8. A 3.00 mol/kg NaCl solution has a density of 1.12 g/mL. Calculate its molarity. (Hint: start with 1.000 kg of water.)

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